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Optimal sizing of stand-alone photovoltaic/wind/hydrogen hybrid system supplying a desalination unit

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Abstract

The real penetration of renewable energy sources (RES) under varying weather conditions required an optimal sizing of hybrid renewable power generation systems. The present paper proposes a methodology based on iterative technique, to perform the optimal sizing of a stand-alone hybrid photovoltaic/wind/hydrogen system supplying a desalination unit which feeds the area's inhabitants with fresh water. The methodology aims at finding the optimal technical–economic configuration among a set of systems components. Taking into account the models of the main hybrid system components and the hourly meteorological data and load profile for any site located worldwide, the recommended algorithm gives all possible configurations that can completely cover the fresh water requirements of isolated consumers. Afterward, the optimal configuration is predicted on the basis of the minimum initial investment cost. A case study is conducted to analyze a hybrid project, which is designed to supply a seawater desalination unit installed on the Kerkennah islands located in the south of Tunisia.
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Keywords: Hybrid photovoltaic/wind system; Fuel cell/electrolyzer storage system; Desalination unit; Optimal sizing; Techno-economic evaluation

1. Introduction

Today, the world is facing a great challenge for saving its future in terms of providing one of the main requirements of humankind: Energy. Nowadays, the majority of the consumed energy all around the world comes from conventional fossil fuels like coal, natural gas, crude oil, etc. (Dursun et al., 2012). Due to the limited reserves of fossil fuels, their rising prices, their polluting of nature and the increasing worry over global warming, the interest in renewable energy sources (RES) has significantly increased (Nasiraghdam and Jadid, 2012; Li et al., 2013).

Compared to available renewable energy options, solar and wind are considered as the most preferred RES for their availability and inexhaustibility (Erdinc and Uzunoglu, 2011; Askarzadeh, 2013). However, a common drawback of these renewable sources is their unpredictable nature and dependence on weather and climatic changes. This issue causes the fact that renewable power production may not totally satisfy the power demand of the load at each instant. To solve this problem, an energy storage unit is usually required to reduce power production fluctuations. Traditionally, batteries have been used in this regard. The short life cycle, the high-cost maintenance and the environmentally unfriendly content of this storage system are the constraining factors for its use in stand-alone applications (Spyrou and Anagnostopoulos, 2010). Recent researches have focused on using fuel cell

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Nomenclature

A_{pv}	surface of PV generator (m^2)
C_{EL}	unit cost of the electrolyzer (\$/kW)
C_{FC}	unit cost of the fuel cell (\$/kW)
C_{Inv}	unit cost of the inverter (\$/kW)
$Cons_{elec}$	average electrical energy consumption to produce one m^3 of fresh water (kW h)
C_{PV}	unit cost of the photovoltaic generator (\$/kW)
$C_{r\cdot hyd}$	unit cost of the hydrogen tank (\$/kg)
C_W	unit cost of the wind system (\$/kW)
G_{ir}	global solar irradiation (W/m^2)
M_{HT}	margin coefficient
N_{da}	number of autonomy days
$Need_{FW}$	hourly average fresh water consumption in July (m^3/h)
N_{oct}	nominal cell operating temperature (°C)
$P_{EL}(t)$	electrolyzer input power (kW) during step Δt
$P_{FC}(t)$	fuel cell output power (kW) during step Δt
P_{Tank}	equivalent power of the fresh water tank content (kW)
$P_{Tank\cdot C}$	equivalent power of the fresh water tank capacity (kW)
P_{Wr}	wind turbine rated power (kW)
T_a	ambient temperature (°C)
T_c	solar cell temperature (°C)
T_{ref}	reference solar cell temperature (°C)
V_C	cut-in wind speed (m/s)
V_F	cut-off wind speed (m/s)
V_R	rated wind speed (m/s)
V_{Tank}	fresh water tank content (m^3)
$V(h)$	wind speed at hub height h (m/s)
$V(h_{ref})$	speed measured at the reference height h_{ref} (m/s)
α	power law exponent
α_{EL}	required energy to produce one kilogram of H_2 (kW h/kg)
α_{FC}	produced energy by one kilogram of H_2 (kWh/kg)
ρ_{H_2}	hydrogen density (0.089 kg/m^3)
η_{EL}	electrolyser efficiency
η_{FC}	fuel cell efficiency
Δt	step time (1 h)

(FC)/electrolyzer (EL) as the storage device due to its reliability and its environmentally-friendly operation (Li et al., 2009; Andrews and Shabani, 2012; Maleki and Askarzadeh, 2014).

In order to efficiently and economically employ the RES, an optimum sizing is quite necessary. The optimal sizing of a hybrid system allows to endorse the lowest investment with a reasonable and full use of the RES, so that the hybrid system can work at optimum conditions in terms of investment and system power reliability requirement. In this context, various optimization techniques for hybrid systems sizing have been reported in the literature.

Tina et al. (2006) presented a probabilistic approach based on the convolution technique to assess the long-term performance of a hybrid solar–wind power system for both stand-alone and grid-connected applications. The disadvantage of this probabilistic approach is that it cannot represent the dynamic changing performance of the hybrid system. A methodology of sizing optimization of a stand-alone hybrid wind/photovoltaic (PV)/diesel energy system has been proposed by Belfkira et al. (2011). This approach makes use of a deterministic algorithm to suggest the number of PV panels, wind turbines (WTs) and diesel system so that the total annual cost of the hybrid system subject be minimized while guaranteeing the availability of the energy. Belmili et al. (2014) present an iterative optimization technique following the loss of power supply probability (LPSP) model for a hybrid solar/wind system. The obtained optimum configuration ensures a reliable power supply with the lowest investment. A methodology for optimal sizing of stand-alone hybrid system based on genetic algorithms has been performed by Koutroulis et al. (2006). The purpose of the proposed methodology is to suggest, among a list of commercially available system devices, the optimal number and type of units ensuring that the 20-year round total system cost is minimized subject to the constraint that the load energy requirements are completely covered. Dufo-López et al. (2011) present a multi-objective optimization for a stand-alone PV–wind–diesel system with batteries storage. The objectives to be minimized are the levelized cost of energy (LCOE) and the equivalent carbon dioxide (CO_2) life cycle emissions (LCE).

In the present work, a new optimal sizing algorithm that utilizes the iterative optimization technique is developed. This optimization approach determines the optimum configuration of a stand-alone PV/wind/hydrogen system supplying a desalination unit that would guarantee a reliable energy supply with the lowest initial investment cost for any site. There are six sizing parameters which can be obtained by this algorithm: the rated power of wind system, the number of PV panels, the rated power of FC, the rated power of EL, the capacity of hydrogen tank and the size of desalination unit.

The paper is organized as follows: Section 2 describes the hybrid system and indicates the relationship between the different components. Section 3 discusses the modeling of the hybrid system components. The proposed optimal sizing algorithm is explained in details in Section 4. The simulation results are discussed in Section 5, and finally, Section 6 establishes the conclusion.

2. Installation description

The studied system is a hybrid installation supplying an isolated load as shown in Fig. 1. This installation consists of PV generator, wind generator, an FC coupled to an EL, converters DC/DC and an inverter which is used to interface the DC voltage to the consumer load AC requirements. The installation includes a fresh water tank that is fed by a desalination unit to provide autonomy to the area for a determined time period. The PV module and the WT work together to meet the load demand. If the total electric energy generated by the RES is greater than the load, the excess electric power is used to operate the EL and produce hydrogen. When power shortage takes place, stored hydrogen converts into electrical energy by using FC which is considered as a secondary power source.

3. Models of system components

Modeling is an essential step toward a good assessment of the hybrid system performance. Indeed, the models of the main system components are developed in the following section.

3.1. PV generator model

The model of the PV generator is slightly complicated by the fact that between the illumination and the temperature, there is a non-linear dependence. In this paper, a simplified simulation model is used to estimate the hourly output power of the PV generator. It is given by (Belmili et al., 2014; Kaabeche et al., 2011; Diafa et al., 2007):

$$P_{PV} = \eta_{pv} A_{pv} G_{ir} \tag{1}$$

The PV generator efficiency is expressed as:

$$\eta_{pv} = \eta_r [1 - \beta (T_c - T_{ref})] \tag{2}$$

T_c is calculated using the NOCT (Normal Operating Cell Temperature), it can be defined as follows:

$$T_c = T_a + G_{ir} \left(\frac{N_{oct} - 20}{800} \right) \tag{3}$$

These different parameters are obtained from the PV module manufacturers.

3.2. Wind turbine system model

The hourly output power of wind generator at a particular location depends on the hourly wind speed at the hub height and the output characteristic of the wind generator. The hourly wind speed at the hub height can be described by using power-law equation (Belfkira et al., 2011; Kaabeche and Ibtiouen, 2014):

$$V(h) = V(h_{ref}) \left(\frac{h}{h_{ref}} \right)^{\alpha} \tag{4}$$

The α value fluctuates from less than 0.1 for very flat land, water or ice to more than 0.25 for heavily forested landscapes.

Depending on this wind speed, the model used to calculate the output power of a WT "$P_W(t)$" can be expressed by (Belmili et al., 2014; Kaabeche et al., 2011):

$$P_W(t) = \begin{cases} P_{Wr} \left[\frac{V^2 - V_C^2}{V_R^2 - V_C^2} \right], & V_C \leqslant V \leqslant V_R \\ P_{W \cdot r} & V_R \leqslant V \leqslant V_F \\ 0 & \text{Otherwise} \end{cases} \tag{5}$$

3.3. Storage system model

There are several different energy storage alternatives like water pumping, flywheels, batteries and hydrogen storage. In this study, the hydrogen storage system is used as it requires less maintenance, it is quieter and it has a better durability in difficult conditions. The hydrogen storage system is composed of FC, EL and hydrogen storage tank (Erdinc and Uzunoglu, 2012).

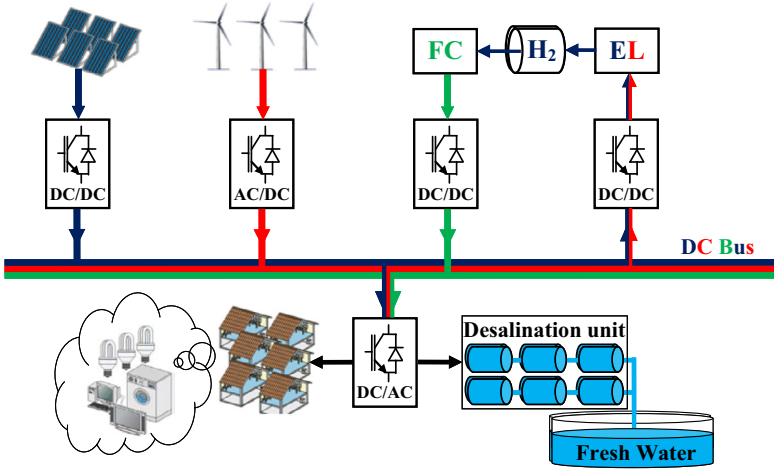


Fig. 1. Schematic diagram of hybrid PV/wind/hydrogen system.

3.3.1. Electrolyzer model

A water EL consists of several cells connected in series. Two electrodes of the EL are separated by an aqueous electrolyte or solid polymer electrolyte. During periods of excess electrical energy of PV and wind sources, the EL uses the excess electricity to decompose water into hydrogen and oxygen (Li et al., 2009). The produced hydrogen can be quantified by the following equation (Dufo-Lopez et al., 2009):

$$m_{H_2\cdot EL}(\Delta t) = \begin{cases} \frac{P_{EL}(t)\cdot\Delta t\cdot\eta_{EL}}{\alpha_{EL}} & \text{if } P_{EL}(t) > 0 \\ 0 & \text{if } P_{EL}(t) = 0 \end{cases} \quad (6)$$

With the required energy to produce one kilogram of H_2 "α_{EL}" is equal to 39.44 kW h/kg (Tebibel and Labed, 2014).

3.3.2. Hydrogen tank model

The hydrogen produced by the EL is stored in the appropriate gas tank at high pressure P_{H_2}. The capacity of the hydrogen tank can be estimated by the following equation (Tebibel and Labed, 2014):

$$V_{H_2} = \frac{m_{H_2\cdot EL} M_{HT}}{P_{H_2}\rho_{H_2}} \quad (7)$$

The margin coefficient M_{HT} is selected from the interval [1.25–3] for accommodating the seasonal fluctuation and the initial hydrogen storage (Castaneda et al., 2013).

3.3.3. Fuel cell model

The proton exchange membrane fuel cell (PEMFC) is composed of several FCs connected in series. The fundamental structure of a PEMFC can be described as two electrodes (anode and cathode) separated by a solid membrane (Li et al., 2009). When the RES are insufficient to supply the load, the FC uses the stored hydrogen to produce the required energy. The amount of needed hydrogen to produce the missing power is expressed by the following equation:

$$m_{H_2\cdot FC}(\Delta t) = \begin{cases} \frac{P_{FC}(t)\cdot\Delta t}{\eta_{FC}\cdot\alpha_{FC}} & \text{if } P_{FC}(t) > 0 \\ 0 & \text{if } P_{FC}(t) = 0 \end{cases} \quad (8)$$

The produced energy by one kilogram of H_2 is $\alpha_{FC} = 33.33$ kW h/kg (Vasallo et al., 2013).

4. Optimal sizing algorithm of a hybrid renewable energy system

The main objective of the sizing algorithm is to determine the optimal configuration of the hybrid system (size of PV generator, wind generator, FC, EL, capacity of the hydrogen tank and size of desalination unit) to desalinate seawater and feed the area's inhabitants with the lowest cost. Several methods for sizing hybrid energy system were discussed in the introduction. In this study, a new techno-economic approach is developed. This algorithm gives all possible configurations of the hybrid system that can supply the load profile demand of any site located worldwide. The optimal solution that has the minimum cost is achieved by applying the economic approach.

4.1. Assumptions and input data of algorithm

Whatever the studied site, the input data set of this algorithm is:

- The hourly data of solar irradiation, wind speed and load demand variations for one year.
- Specifications of the different hybrid system components.
- $P_{Wr\cdot min}$ and $P_{Wr\cdot max}$: the lower and higher limits of the variation interval of wind generator rated power.
- $N_{PPV\cdot min}$ and $N_{PPV\cdot max}$: the lower and higher limits of PV panels number.
- ΔN_{PPV} and ΔP_W: the variation step of PV panels number and wind generator power, respectively.
- Δt: the simulation step.
- $V_{Tank\cdot C}$ and $V_{Tank\cdot i}$: the capacity and the initial state of the fresh water tank, respectively.
- $P_{DES\cdot inst}$: the initial size of the desalination unit.

Due to the commercial unavailability of all the WTs required sizes, a sub-algorithm of a wind generator selection is developed to study the most possible solutions. The wind generator may be constituted by a single WT or by a combination of identical WTs. The sub algorithm is described with the flowchart presented in Fig. 2.

The wind turbines are classified according to their powers in increasing order. The sub-algorithm starts by defining the expected power of the wind generator "P_W". If P_W is found in the list of commercially available wind turbines, it will be chosen to be installed. Otherwise, the following wind turbine in the list "P_{Ws}" is automatically installed providing less power than the next expected one ($P_{Ws} < P_W + \Delta P_W$). If no above case is valid, two identical wind turbines ($P_{Wq} = P_W/2$) that provide the expected power "P_W" are installed. If this turbine is not found within the list or it is used by the previous solutions, two other identical wind turbines "P_{Wf}" which are classified in the following rank in the commercial list of wind turbines are studied and so on until finding the solution. The sub-algorithm runs to a new expected power of the wind generator if the total power of both wind turbines exceeds the higher limit of wind generator interval or if the commercial list is studied without finding a solution.

4.2. Technical developed algorithm analysis

The present section is designed to identify all combinations of the hybrid system which guarantee the energy autonomy of consumers for a selected time period "Time". Fig. 3 represents the flowchart of the proposed algorithm.

The developed algorithm consists of the following steps:

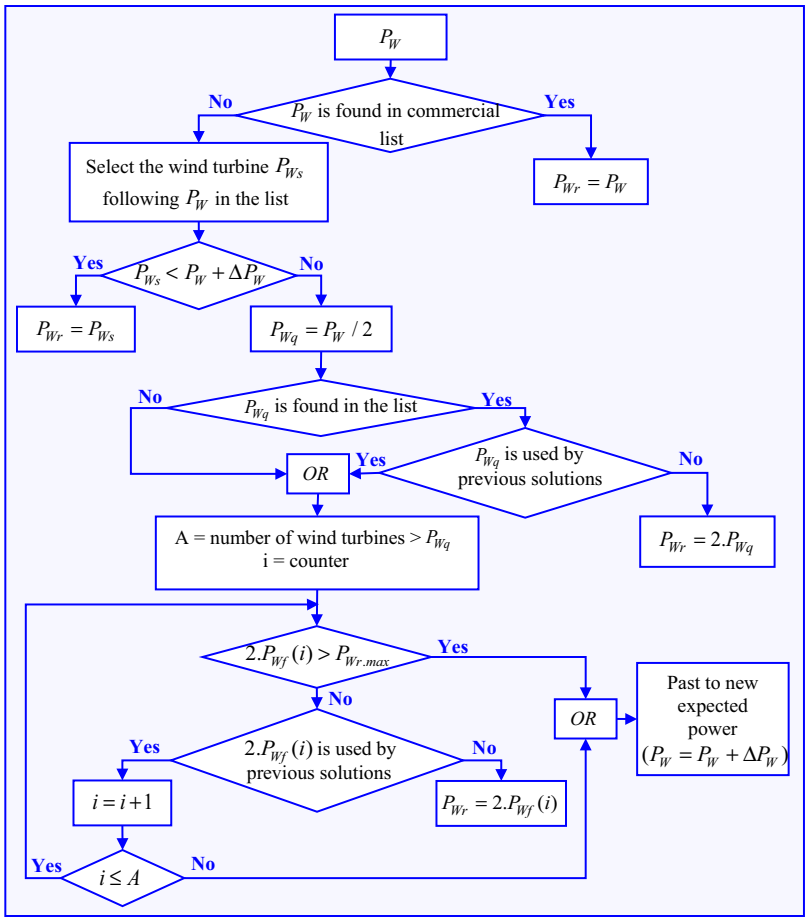


Fig. 2. Sub-algorithm of wind generator selection.

Step 1. Select the wind generator rated power "P_{Wr}" within a specific numerical range $[P_{Wr\cdot min}, P_{Wr\cdot max}]$.
Step 2. Define the number of PV panels "N_{PPV}" of a peak power "$P_{PV\cdot p}$" each.
Step 3. In every time-step of the simulation, the produced energies by the wind generator "$P_W(t)$" and the PV generator "$P_{PV}(t)$" are computed using their mathematical models, taking into account the existing wind speed, the available solar irradiation, the selected WT characteristics as well as the parameters of the PV panel. The total produced power by RES is given by the following relationship:

$$P_{RES}(t) = P_W(t) + P_{PV}(t) \tag{9}$$

The following algorithm consists of two main parts.

Step 4. The first part (Fig. 3a) allows calculating the FC installed power "$P_{FC\cdot inst}$" and the EL installed power "$P_{EL\cdot inst}$". It consists of the subsequent steps:

Step 4.1. In this part, the hourly water demand represents the load. The correlation between the hourly water demand "$V_D(m^3)$" and the corresponding power for desalination "$P_D(t)$" is given by:

$$P_D(t) = V_D Cons_{elec} \tag{10}$$

In general, reverse-osmosis desalination units consume between 3 and 4 kW h to produce one m^3 of fresh water (Fritzmann et al., 2007). In the present study, an average electrical energy consumption of 3.5 kW h/m^3 is used.

Step 4.2. Compare the generated power by the PV/wind system with the power demand "$P_D(t)/\eta_{inv}$".

$$\delta P = P_D(t)/\eta_{inv} - P_{RES}(t) \tag{11}$$

If "$P_{RES}(t)$" is greater than the load demand "$P_D(t)/\eta_{inv}$". In this case, the excess energy is stored in table "T_{EL}". Otherwise, in the case of energy deficiency "$\delta P > 0$", the difference is stored in table "T_{FC}".

Step 4.3. After the loop execution during one year "Time = 8760 h", "$P_{EL\cdot av}$" and "$P_{FC\cdot av}$" are computed by the following equations:

$$P_{EL\cdot av} = mean(T_{EL}) \tag{12}$$

$$P_{FC\cdot av} = mean(T_{FC}) \tag{13}$$

Among a list of commercially available EL devices, the suitable EL is chosen. If "$P_{EL\cdot av}$" is existing within the list; therefore, "$P_{EL\cdot inst} = P_{EL\cdot av}$". Otherwise, the EL, that has a

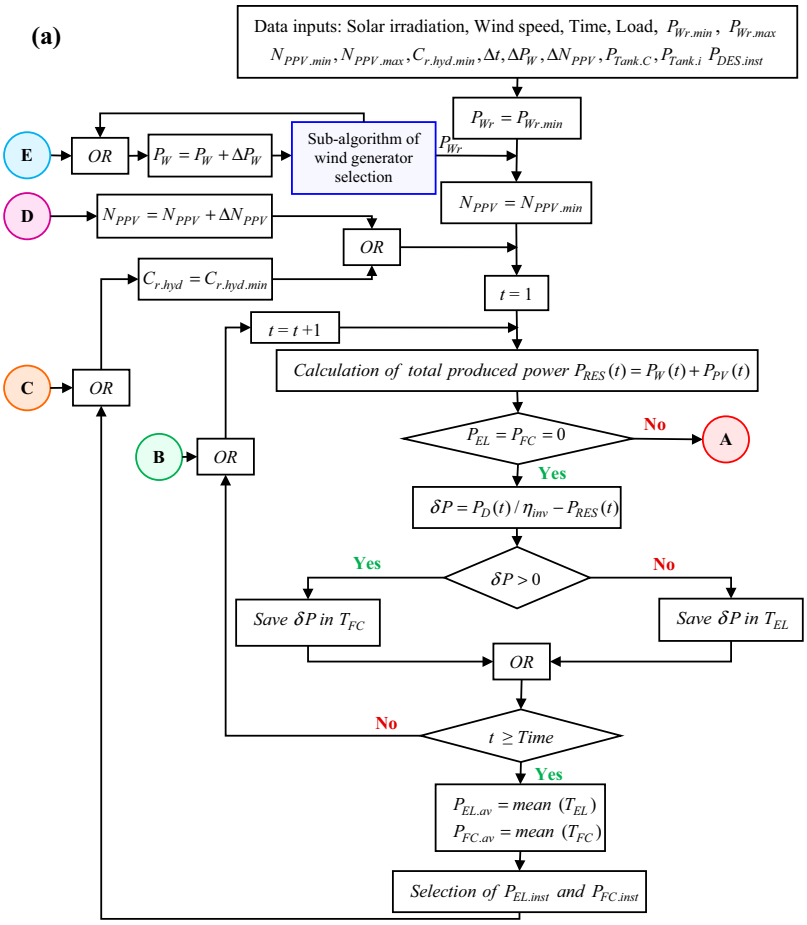


Fig. 3. Flow chart of the optimal sizing algorithm. (a) Primary selection of EL and FC installed power and (b) adjustment and evaluation of the studied configuration.

power greater than calculated, is selected to be installed. The same steps are treated for the FC and we get "$P_{EL\cdot inst}$" and "$P_{FC\cdot inst}$".

The maximum powers of the EL and the FC are given by:

$$P_{EL\cdot max} = max(T_{EL}) \quad (14)$$

$$P_{FC\cdot max} = max(T_{FC}) \quad (15)$$

Step 5. In the second part (Fig. 3b), the proposed combination (P_{Wr}, N_{PPV}, $P_{EL\cdot inst}$, $P_{FC\cdot inst}$, $P_{DES\cdot inst}$) is tested to assess its technical performance. This part includes the following steps:

Step 5.1. Define the initial capacity of the hydrogen tank "$C_{r\cdot hyd}$". It is chosen to be zero in this study.

Step 5.2. Return to Step 3 to calculate the generated power from the PV/wind system "$P_{RES}(t)$", then proceed to (A).

Step 5.3. The water consumers are always being fed through the tank. The desalination unit operates in order to replace the consumed water of the tank. In this case, the new load expression can be defined as follows:

$$V_{DW} = V_{Tank\cdot C} - V_{Tank} \quad (16)$$

The desalination power equivalent of the required fresh water to fill the tank is calculated as:

$$P_{DW} = (V_{Tank\cdot C} - V_{Tank}) \cdot Cons_{elec} \quad (17)$$

$$P_{DW} = P_{Tank\cdot C} - P_{Tank} \quad (18)$$

Step 5.4. Compare the produced power by renewable energy sources "$P_{RES}(t)$" with "$P_{DW}(t)/\eta_{inv}$".

$$\delta P_W = P_{DW}(t)/\eta_{inv} - P_{RES}(t) \quad (19)$$

Step 5.5. If the total produced power by RES is greater than the required power to completely fill the water tank "$P_{RES}(t) > P_{DW}(t)/\eta_{inv}$", the consumed power by the desalination unit "$P_{DES}(t)$" is equal to the minimum between the required power to completely fill the tank "$P_{DW}(t)/\eta_{inv}$" and the unit installed power "$P_{DES\cdot inst}/\eta_{inv}$", to ensure the operation of the desalination unit with a power less than or equal to the rated value. In this case, there is always an excess energy that is absorbed by the EL to be converted into hydrogen. The EL power is within the minimum range of the following powers; the excess power "$P_{exc}(t)$" (Eq. (20)) and the EL rated power "$P_{EL\cdot inst}$". The new state of charge of hydrogen

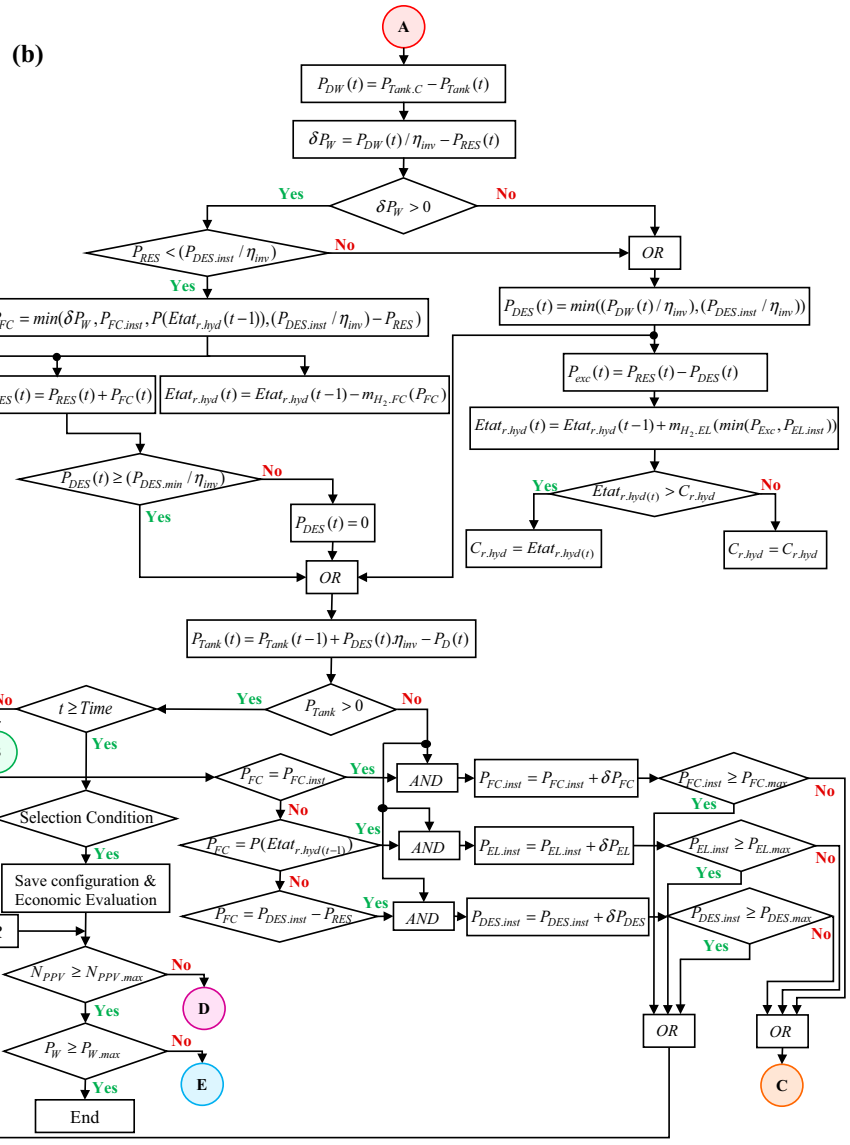


Fig 3. *(continued)*

tank "$Etat_{r\cdot hyd}(t)$" is calculated like Eq. (21). If the amount of stored hydrogen "$Etat_{r\cdot hyd}(t)$" is higher than the hydrogen tank capacity "$C_{r\cdot hyd}$", the hydrogen tank size is increased by a given quantity to become able to store this excess, thus "$C_{r\cdot hyd} = Etat_{r\cdot hyd}(t)$". Otherwise, the hydrogen tank keeps the same capacity. Afterward, proceed to Step 5.7.

$$P_{exc}(t) = P_{RES}(t) - P_{DES}(t) \tag{20}$$

$$Etat_{r\cdot hyd}(t) = Etat_{r\cdot hyd}(t-1) + m_{H_2\cdot EL}(min(P_{exc}, P_{inst\cdot EL})) \tag{21}$$

Step 5.6. When the energy demand is greater than the available energy generated by the PV/wind system "$P_{DW}(t)/\eta_{inv} > P_{RES}(t)$", it can have two cases:

- *First case:* $P_{RES}(t) > P_{DES\cdot inst}/\eta_{inv}$, there exists an energy deficit, but the desalination unit cannot meet the lack of energy since it operates at its rated power "$P_{DES\cdot inst}/\eta_{inv}$" which is less than the produced energy by RES "$P_{RES}(t)$" and the required energy by the load demand "$P_{DW}(t)/\eta_{inv}$". In this case, there is always an unused electrical energy that will be stored by the EL as hydrogen.
- *Second case:* $P_{RES}(t) < P_{DES\cdot inst}/\eta_{inv}$, in this situation, the energy deficiency is covered by the FC system. The latter "$P_{FC}(t)$" operates with the minimum of the following powers; the missing power to fill the fresh water tank "$\delta P_W = P_{DW}(t)/\eta_{inv} - P_{RES}(t)$", the FC rated power "$P_{FC\cdot inst}$", the missing power to operate the unit at its rated power "$P_{DES\cdot inst}/\eta_{inv}-$

$P_{RES}(t)$" and the stored power as hydrogen "$P(Etat_{r\cdot hyd}(t-1))$". The new state of charge of hydrogen tank at hour "t" can be calculated by:

$$Etat_{r\cdot hyd}(t) = Etat_{r\cdot hyd}(t-1) - m_{H_2\cdot FC}(P_{FC}) \quad (22)$$

If the sum of the produced power by RES and the FC is adequate to operate the desalination unit at its minimum power, the consumed power by the desalination unit is expressed by the following equation:

$$P_{DES}(t) = P_{RES}(t) + P_{FC}(t) \quad (23)$$

If this is not the case "$(P_{RES}(t) + P_{FC}(t)) < (P_{DES\cdot min}/\eta_{inv})$", the desalination unit does not operate.

Step 5.7. Calculate the fill level of fresh water tank after the desalination and the inhabitants' consumption by:

$$P_{Tank} = P_{Tank} + P_{DES}\cdot\eta_{inv} - P_D(t) \quad (24)$$

Step 5.8. If the fresh water tank still contains water, a new time point is analyzed (return by (B) and go to Step 5.2).

Step 5.9. If the fresh water tank is empty; therefore, the proposed combination cannot satisfy the water demand and some modifications are quite necessary:

- If "$P_{Tank} < 0$" and "$P_{FC}(t) = P_{FC\cdot inst}$" (i.e. there is hydrogen in the storing tank, but the FC installed power is low); thus, the FC size is increased by a given variation step.

$$P_{FC\cdot inst} = P_{FC\cdot inst} + \delta P_{FC} \quad (25)$$

- If "$P_{Tank} < 0$" and "$P_{FC}(t) = P(Etat_{r\cdot hyd}(t))$" (i.e. the amount of stored hydrogen in the tank is not enough), thus the EL installed power is increased.

$$P_{EL\cdot inst} = P_{EL\cdot inst} + \delta P_{EL} \quad (26)$$

- If "$P_{Tank} < 0$" and "$P_{FC}(t) = P_{DES\cdot inst}/\eta_{inv} - P_{RES}(t)$" (i.e. there is a stored hydrogen in the tank, but the desalination unit, with its rated power, cannot satisfy the load demand); thus, the desalination unit installed power is increased.

$$P_{DES\cdot inst} = P_{DES\cdot inst} + \delta P_{DES} \quad (27)$$

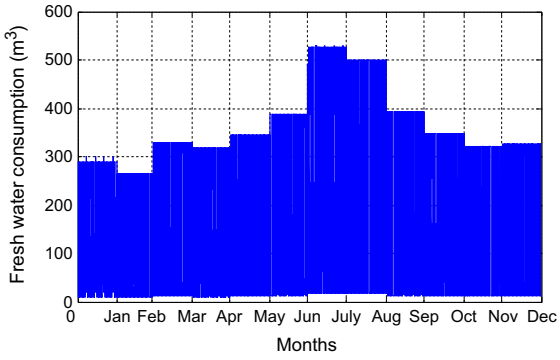


Fig. 4. Hourly fresh water consumption.

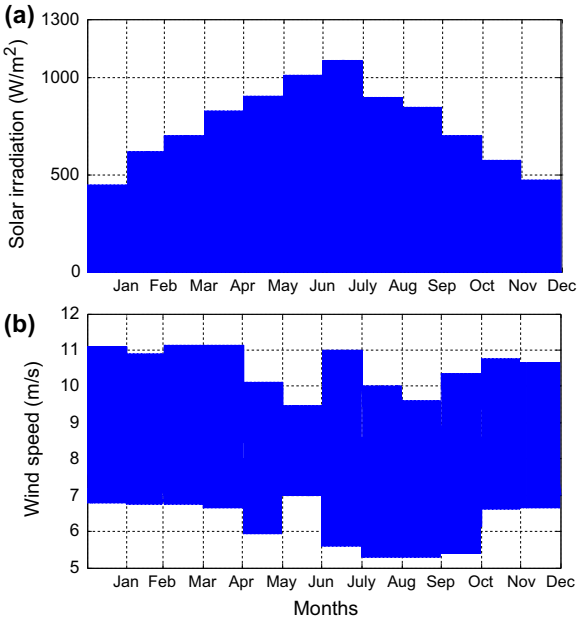


Fig. 5. Meteorological conditions for the studied site: (a) hourly solar irradiation and (b) hourly wind speed.

After the amendment of the configuration with one of the above three cases, the new combination is retested (return by (C) and proceed to Step 5 with $t = 1$).

Powers "$P_{EL\cdot inst}$", "$P_{FC\cdot inst}$" and "$P_{DES\cdot inst}$" should not exceed the maximum power limits of EL "$P_{EL\cdot max}$", FC "$P_{FC\cdot max}$", and the desalination unit "$P_{DES\cdot max}$", respectively. Otherwise, go to Step 5.11.

Step 5.10. After the algorithm simulation during one year "Time = 8760 h", (i.e. when t = Time), a hybrid

Table 1
Costs for the system components.

Component	Cost
PV array	2000 ($/kW) (Ma et al., 2014; Rahimi et al., 2014)
Wind turbine	1500 ($/kW) (Nasiraghdam and Jadid, 2012; Genc et al., 2012)
Fuel cell	2000 ($/kW) (Andrews and Shabani, 2012; Rahimi et al., 2014)
Electrolyzer	1500 ($/kW) (Andrews and Shabani, 2012; Rahimi et al., 2014)
Hydrogen tank	500 ($/kg) (Andrews and Shabani, 2012; Rahimi et al., 2014)
Inverter	700 ($/kW) (Li et al., 2013; Kaabeche and Ibtiouen, 2014; Tebibel and Labed, 2014)

Table 2
Specifications of the wind turbines.

Type	Rated power (kW)	Cut-in wind speed (m/s)	Rated wind speed (m/s)	Cut-off wind speed (m/s)	Hub height (m)
Hummer H12.0-50KW Wind Turbine	50	3	12	25	18
Siva 100	100	3	10	25	30
Siva 200	200	4	10	25	30
Siva 225/40	225	4	13	25	30
Direct Drive Wind TurbinesDW52/54	250	3	8/7.5	25	50
PalmTree 300	300	3	12	20	32
DW52/54-500kW	500	3	10	25	52
Turbowinds T600-48	600	3	12.5	25	50
Garuda 700.54 700kW	700	3	12.5	24	73
SUREnergy SURU54 177-750	750	3	11.5	25	60
DDIS60-800kw wind turbine	800	3	10.3	20	60
LTW77-850	850	3	11	25	65
PowerWind 56-900kW	900	3	12.5	25	59
Alizeo 1000/64 Softwind	1000	3.5	11	15	50
Shanghai Electric W1250/70	1250	3.5	11.5	23	55
Acciona AW-1500/77	1500	3.5	11	25	77
GE 1.6-100	1600	3.5	11	25	80
Ghodawat G1650	1650	3.5	11.5	20	78
GARUDA 1700.84	1700	3.5	10.76	24	80.5
Vestas V100/1800	1800	4	12	20	80

Table 3
Specifications of the PV module.

Parameters	Value
Type	Istar Solar® IS4000P
Nominal peak power (W)	290
Reference efficiency (%)	15.4
Optimum operating voltage (V_{mp})	37.6
Optimum operating current (I_{mp})	7.7
Open circuit voltage (V)	44.4
Short-circuit current (A)	8.52
Dimensions	1960 × 985 × 42 mm

system composed of a wind generator "P_{Wr}", PV panels "N_{PPV}", an EL "$P_{EL\cdot inst}$", a FC "$P_{FC\cdot inst}$", a hydrogen storage tank "$C_{r\cdot hyd}$" and a desalination unit "$P_{DES\cdot inst}$" is obtained. The obtained configuration is able to satisfy the load demand and ensure the energy autonomy of the system. In order not to retain an oversized system, the obtained configuration must satisfy the following condition:

- According to the state variation of the hydrogen tank during one year, "$Etat_{r\cdot hyd}(t)$" must reach its minimum level at least once.

If the system satisfies the condition, the configuration is saved to calculate its installation cost. Afterward, start a new configuration as shown in Step 5.11. In the opposite case, if the condition is not satisfied, proceed to Step 5.11.

Step 5.11. Start a new configuration by adding "ΔN_{PPV}" to the existing number of PV panels (go to Step 2 (D)). If the maximum available number of PV panels "$N_{PPV\cdot max}$" is reached, a new WT rated power is selected instead of the first one, while the algorithm restarts from Step 1 (return via (E)).

Using the developed algorithm, all possible configurations of the stand-alone hybrid system are studied. Several combinations, which satisfy the system power reliability requirements, can be obtained. The optimum configuration is subsequently predicted on the basis of the minimum initial cost criterion.

4.3. Economic developed algorithm analysis

As already discussed in previous work, the initial investment cost is related to the purchase cost added to installation cost. It can be computed from the relation (Belmili et al., 2014; Kaabeche et al., 2011; Tebibel and Labed, 2014):

$$C_i = P_{Wr}C_W + N_{PPV}P_{PV\cdot p}C_{PV} + P_{EL\cdot inst}C_{EL} + P_{FC\cdot inst}C_{FC} + P_{Inv}C_{Inv} + C_{r\cdot hyd}C_{kg\cdot H_2} \quad (28)$$

The unit price of each component (PV generator, WT, FC, EL, hydrogen tank and inverter) in this study is shown in Table 1.

5. Results and discussion

In order to highlight the recommended sizing methodology, a case study is conducted to investigate a stand-alone PV/wind/hydrogen hybrid system which is intended to supply a seawater desalination unit installed on the Kerkennah islands located in the south of Tunisia. The desalination unit feeds a population of 14,400 inhabitants with fresh water in winter. During the summer months, the fresh water consumption increases because of tourism, it varies according to the season. The water consumption has an average annual value of 193.6 m^3 and a maximum of

530 m^3 as it is depicted in Fig. 4 (SONEDE Kerkennah, 2014).

Hourly data of solar irradiation and wind speed of the studied site during one year are plotted in Fig. 5. The data have been extracted from the NASA satellite after entering the geographical coordinates of the Kerkennah islands (Latitude: 34.739°, longitude: 11.188°) (NASA Surface meteorology and Solar Energy, 2014). The wind speed was measured at a 10 m height which is considered as the reference height for the selected site (cf. Eq. (4)).

The technical characteristics of the WTs and PV panel used in this study are listed in Tables 2 and 3, respectively. Table 2 shows the WTs list the most adapted to wind potential of the selected site (hourly wind speed varies between 5.3 m/s and 11.15 m/s with an average annual value of 8.35 m/s).

For finding the optimum design of the hybrid system, the proposed algorithm has been implemented using MATLAB software. The parameters input setting of the algorithm is as follows:

$P_{Wr\cdot min} = 0$ kW and $P_{Wr\cdot max} = 1800$ kW

$N_{PPV\cdot min} = 0$ and $N_{PPV\cdot max} = 3600$ PV panels

$\Delta N_{PPV} = 50$ PV panels, $\Delta P_W = 100$ kW and $\Delta t = 1$ h

The installed fresh water tank ensures three days of autonomy in July, the capacity of the fresh water tank is given by the following equation:

$$V_{Tank\cdot C} = 24 \cdot N_{da} \cdot Need_{FW} \quad (29)$$

The simulation results achieved by the sizing algorithm are demonstrated in Fig. 6. These figures show all the possible system configurations that meet the load demand for the selected site under given meteorological conditions. These curves represent the complementarity between the two RES, a maximum number of PV modules corresponding to the lower rated power of wind generator and vice versa.

Fig. 6a illustrates the variation of the FC power as a function of the PV panels number for different wind generator sizes, Fig. 6b shows the distribution of the EL power according to the various hybrid system configurations and Fig. 6c presents the variation of the hydrogen tank capacity of the above presented solutions.

For each pair (N_{PPV} and P_{Wr}), these three curves give the corresponding combination of $P_{FC\cdot inst}$ (Fig. 6a), $P_{EL\cdot inst}$ (Fig. 6b) and hydrogen tank capacity (Fig. 6c) to represent a configuration that ensures the total autonomy of the system for the examined time period. Keep in mind that, according to the simulation results, the desalination unit installed power is found to be 1050 kW for all obtained configurations. Fig. 6a reveals that for the curve of $P_{Wr} = 1600$ kW, the FC maintains the same size 500 kW. While for the other solutions, it varies between 475 and 550 kW.

In Fig. 6b, for most of the wind generators investigated, the EL rated power undergoes a reduction when the PV panels number increases and subsequently it stabilizes, which is more obvious up to the curve of $P_{Wr} = 1250$ kW.

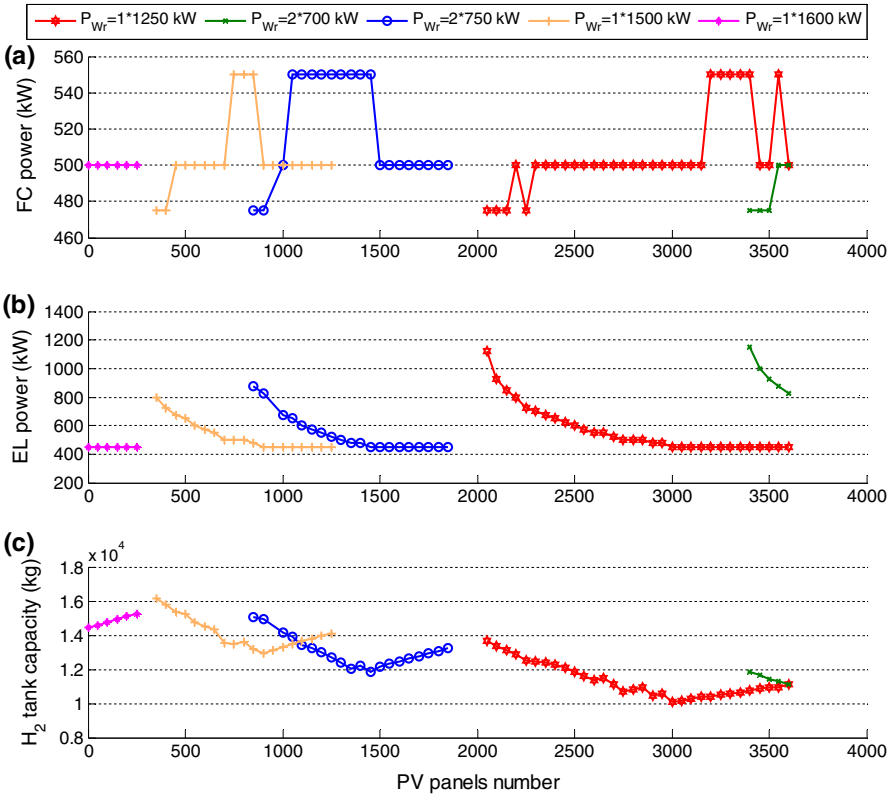


Fig. 6. Distribution of various system configurations according to: (a) fuel cell power, (b) electrolyzer power and (c) hydrogen tank capacity.

On top of this, according to the curves shown in Fig. 6b and c, one may note that the reduction of the EL rated power leads to decreasing the hydrogen tank capacity. Whereas, in the case of a constant value of the EL power, a significant increase of the hydrogen tank capacity is encountered due to the increase of the PV panels number.

To find the best design of the stand-alone system configuration, a comparison between the different configurations that have the same sizes of EL and FC is treated. A careful observation into Fig. 6c shows that there are four groups; each one is devoted to a studied wind generator. The fifth wind generator of rated power $P_{Wr} = 2 * 700$ kW does not participate in the comparison since it does not have solutions that have the same size of EL. For the first wind generator $P_{Wr} = 1600$ kW, the PV panels number varies between zero and 250, while the second group varying between 900 and 1250 is devoted to the wind generator of $P_{Wr} = 1500$ kW. For the third wind generator $P_{Wr} = 2 * 750$ kW, the PV panels number varies between 1450 and 1850. Finally, for the fourth group concerning $P_{Wr} = 1250$ kW, the PV panels number varies between 3000 and 3600. By comparing the four groups, one may see that the more the WT capacity and the PV panels number are both moderate (i.e. from the group of $P_{Wr} = 1600$ kW to the group of $P_{Wr} = 1250$ kW), the more the hydrogen tank capacity decreases. The encountered decrease is much sharper while comparing the wind only system ($P_{Wr} = 1600$ kW, $N_{PPV} = 0$ pv panels) with the hybrid system composed of WT of 1250 kW and 3000 PV panels. More specifically, the hydrogen tank capacity drops from 14,439 kg (811.17 m^3) to 10,078 kg (566.17 m^3) while switching from wind-only system (solution 1) to the hybrid system (solution 2). In order to explain the results, the distribution of the excess energy and the hydrogen tank fill level of the two solutions are shown in Figs. 7 and 8, respectively. According to these curves, three main periods can be defined throughout the year; period A (January to May, $0\text{ h} < t < 3624$ h), period B (June to October, $3625\text{ h} < t < 7296$ h) and period C (November and December, $7297\text{ h} < t < 8760$ h). During periods A and C, the winter and the spring provide a high wind speed and a low solar radiation, the excess energy of the wind-only system is greater than the hybrid PV/wind system since the wind is the main renewable source during these periods. Although these two systems have the same size of the EL, the amount of stored hydrogen by the first system is greater than the second one due to the fact that its EL works throughout the period at full capacity while the EL of the second solution only works with variable power over

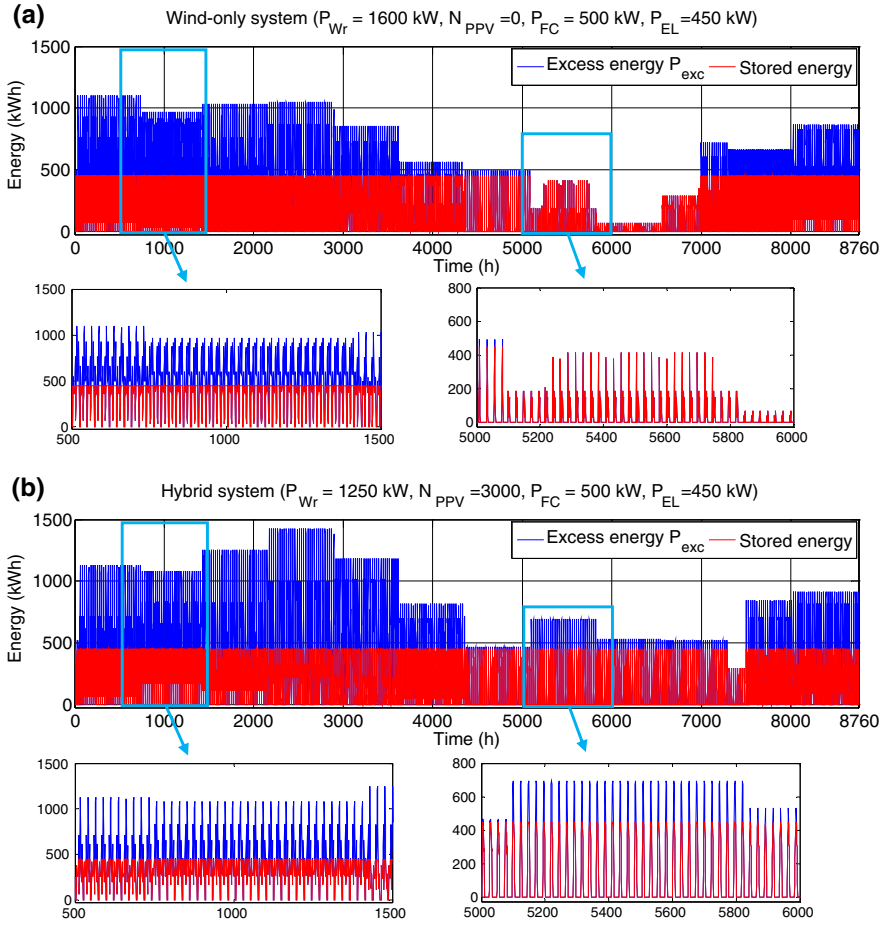


Fig. 7. Distribution of excess energy: (a) first configuration and (b) second configuration.

an intermittent period, as shown by zoom of Fig. 7. During period B, the wind speed is less important, whilst the solar radiation undergoes an increase. With the summer conditions, the PV array reaches its maximum production. Accordingly, the hybrid PV/wind system produces more energy than the wind-only system; therefore, it stores more energy (zoom of Fig. 7). Due to the high water needs in summer season, the electric power consumption is great, the FC is continuously used to supplement the desalination power, as long as there is available hydrogen stored in the tank. Fig. 8 shows that although the hydrogen tank of the wind-only system contains more hydrogen, it is exhausted almost at the same time with the hybrid system tank. That is because of the low wind generator production during the summer. When the storage system fails to compensate the energy deficit, as shown by the zoom of Fig. 8, the desalination unit is powered only by RES and the population is always fed by the fresh water tank.

Based on the above obtained results and according to the weather conditions of the Kerkennah islands which show the complementarity of the solar and the wind sources, the combination of a PV generator with a wind system significantly reduces the hydrogen tank capacity and enhances the system's reliability. Hence, the hybrid generation system is recommended for the present case study.

Fig. 9 shows the initial investment cost distribution of different optimal configurations. As seen from the obtained results, one point with the minimum initial investment cost occurs in each curve of studied wind generator. These configurations are considered as the optimal ones which meet the system reliability requirement with the lowest cost. Moreover, according to Fig. 6a and b, the optimal systems have the same size of FC and EL. A deeper inspection into Fig. 9 shows that a significant investment cost increase is encountered by increasing the hydrogen tank capacity.

According to the previous results, confirming that the moderation of the two renewable sources can allow their good penetration in the studied site and based on the present remarks, the optimal combination of the stand-alone system that can reliably supply electric power to the desalination unit with the lowest initial investment cost is composed of a WT of rated power 1250 kW, 3000 PV panels, EL power of 450 kW, FC power of 500 kW, hydrogen tank capacity of 10,078 kg and a desalination unit installed power of 1050 kW. As indicated in Eq. (7), the volume of the hydrogen tank is calculated taking into account the initial state of hydrogen tank. It is taken equal to 707.7 m^3.

To increase the reliability of the results provided by the sizing algorithm and to validate the technical performance of the optimal configuration, the different curves describing the behavior of the system are presented in Fig. 10.

Due to complementary nature of solar energy and wind energy, both wind and PV generators present the primary source of energy. With reference to Fig. 10a, which shows the hourly total produced power by RES, it can be observed that the solar energy and the wind energy resources can compensate for each other not only for a few days but also for the whole year.

Fig. 10b displays the consumed energy by the desalination unit. Fig. 10c and d presents the EL energy and the FC energy, respectively. As one may see, the desalination unit works most of the time in its rated power. It is powered by the RES or the FC or both. In winter, the energy is produced mostly by wind generator while the remaining is completely guaranteed by the PV generator. This time, the EL that generates hydrogen from excess power production works at full capacity as shown in Fig. 10c. On the other hand, during the summer, the water demand is higher and the provided energy by the WT and the PV generator is

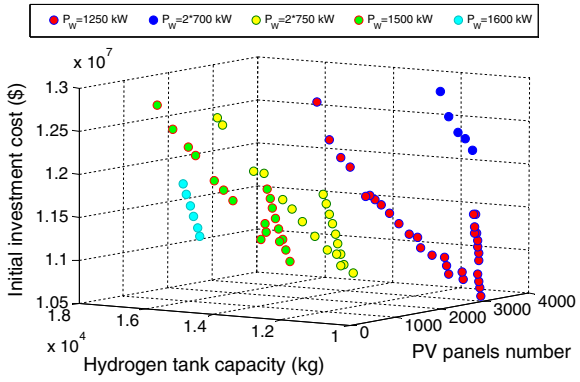


Fig. 9. Initial investment cost distribution of various system configurations.

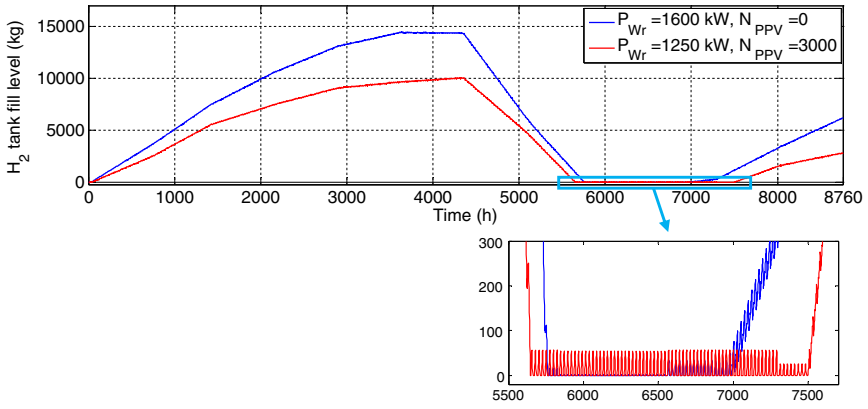


Fig. 8. Hydrogen tank content.

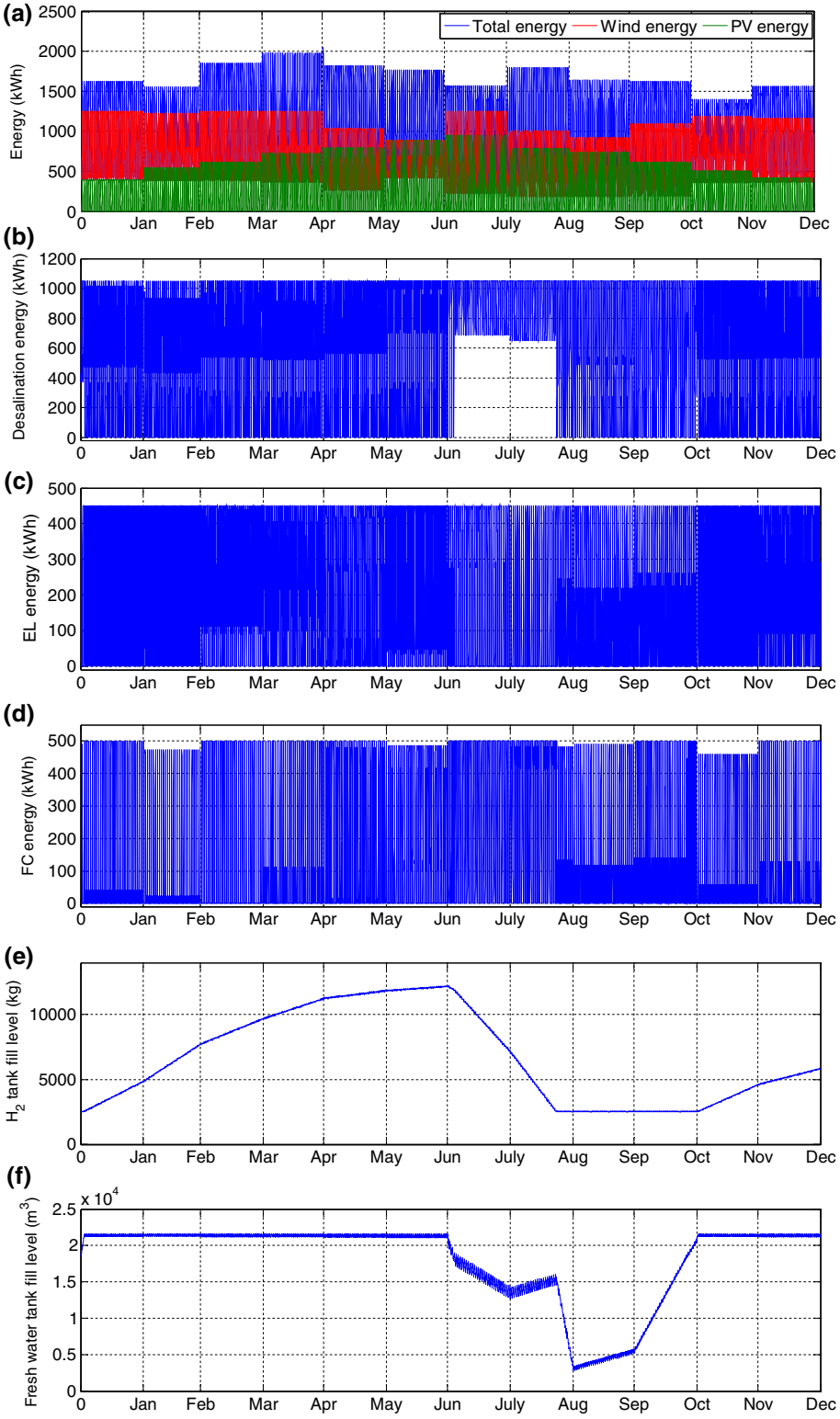


Fig. 10. Plant monitoring of the optimal configuration during one year: (a) produced energies by the RES, (b) desalination energy, (c) EL energy, (d) FC energy, (e) hydrogen tank content and (f) fresh water tank content.

not enough to cover the energy needs of the desalination plant. In this case, the FC is continuously used to supplement the desalination unit power (Fig. 10d). The fill level of hydrogen tank presented in Fig. 10e shows the above results. Furthermore, from Fig. 10e, one may note that the fill level of the hydrogen tank can never exceed its capacity, and can never be below the permissible minimum value (20% of $C_{r\text{-}hyd}$). In the period of the hydrogen tank exhaustion, the water demand is guaranteed and available to the consumers through the fresh water tank.

Fig. 10f highlights the fill level of fresh water tank. With this desalination plant, all water needs of the Kerkennah islands population are satisfied whether in winter or in summer even in periods of low desalination production.

6. Conclusion

An optimal sizing algorithm of stand-alone hybrid PV/wind system with hydrogen storage system, using an iterative optimization technique, is developed in this paper. The system can supply any desalination unit located worldwide that feeds the area's inhabitants with fresh water. The recommended methodology consists of three main parts: the modeling of the hybrid system components, the technical analysis for system reliability evaluation and the economic analysis for system economic profitability evaluation. A set of hybrid system configurations which meet the system reliability requirements are achieved. The configuration with the lowest initial investment cost gives the optimal one.

The developed sizing algorithm is applied to a hybrid project, which is designed to supply a seawater desalination unit installed on the Kerkennah islands located in the south of Tunisia. The weather conditions of the Kerkennah islands represent the best complementarity between the two RES photovoltaic and wind. Based on the results obtained by the developed algorithm, it can be stated that the combination of a PV generator with a wind system significantly can effectively improve energy supply reliability and reduce the energy storage requirements and consequently reduces the system installation cost.

The proposed sizing search method can significantly reduce the computational effort. The simulation results show that this algorithm can size very carefully a hybrid PV/wind system with hydrogen storage supplying a desalination unit. This algorithm allows to adjust the size of WT, PV generator, FC, EL, hydrogen tank and desalination unit with a well defined quantity without estimation such as presented in the previous studies. All obtained configurations guarantee the total autonomy of the system. The hourly energy balance analysis of the lowest cost configuration shows the technical feasibility of the optimal configuration.

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